

Module 3: Functions of a random variable and its expectation

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1 Probability distribution of a function of a discrete random variable

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow \mathbb{R}$ be a random variable (RV) with distribution function (DF) F , probability mass function (pmf) f and support S . Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a given function. Define $Z : \Omega \rightarrow \mathbb{R}$ as

$$Z(\omega) = h(X(\omega)), \omega \in \Omega.$$

Then Z is a RV and it is a function of RV X . Since we are only interested in values of RV X and Z and not the original probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We simply write $X(\omega)$, $\omega \in \Omega$ as X and $Z(\omega)$, $\omega \in \Omega$ as Z . We have

$$F(x) = \mathbb{P}(X \leq x), x \in \mathbb{R}$$

$$f(x) = \mathbb{P}(X = x), x \in \mathbb{R}$$

$\mathbb{P}(X \in S) = 1$ and $\mathbb{P}(X = x) > 0$, $x \in S$.

Define $T = h(S) = \{h(x) : x \in S\}$. Then T is countable. For any set $A \subseteq \mathbb{R}$, define $h^{-1}(A) = \{x \in S : h(x) \in A\}$. For $z \in T$,

$$\mathbb{P}(Z = z) = \mathbb{P}(h(X) = z) = \sum_{x \in S : h(x) = z} \mathbb{P}(X = x) \quad (1)$$

$$= \sum_{x \in h^{-1}\{z\}} \mathbb{P}(X = x) = \sum_{x \in h^{-1}\{z\}} f(x). \quad (2)$$

Thus $\mathbb{P}(Z = z) > 0$ for all $z \in T$ (since $\mathbb{P}(X = x) > 0$, for all $x \in S$) and $\mathbb{P}(Z \in T) = 1$ (since $\mathbb{P}(X \in S) = 1$). Further for any $z \notin T$, $\mathbb{P}(Z = z) = 0$. Thus we have the following theorem

Theorem 1.1. *Let X be a discrete random variable with support S , distribution function F and probability mass function f . Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a given function. Then $Z = h(X)$ is a discrete RV with support $T = \{h(x) : x \in S\}$, PMF*

$$g(z) = \begin{cases} \sum_{x \in h^{-1}(\{z\})} f(x), & \text{if } z \in T \\ 0, & \text{Otherwise.} \end{cases}$$

and DF

$$G(z) = \mathbb{P}(Z \leq z) = \sum_{t \in T : t \leq z} g(t) = \sum_{\{x \in S : h(x) \leq z\}} f(x) = \sum_{x \in h^{-1}((-\infty, z]) \cap S} f(x).$$

In particular when $h : S \rightarrow \mathbb{R}$ is one-one, then

$$g(z) = \begin{cases} f(h^{-1}(z)), & \text{if } z \in T \\ 0, & \text{Otherwise.} \end{cases}$$

Example 1.1. Let X be a discrete RV with PMF

$$f(x) = \begin{cases} \frac{1}{7}, & \text{if } x \in \{-2, -1, 0, 1\} \\ \frac{3}{14}, & \text{if } x \in \{2, 3\} \\ 0, & \text{Otherwise.} \end{cases}$$

Find the PMF and DF of $Y = X^2$.

Solution: Here the support of X is $\{-2, -1, 0, 1, 2, 3\}$. By Theorem 1.1 $Y = X^2$ is a discrete RV with support $T = \{0, 1, 4, 9\}$ and PMF

$$\begin{aligned} g(z) = \mathbb{P}(X^2 = z) &= \begin{cases} \mathbb{P}(X = 0), & \text{if } z = 0 \\ \mathbb{P}(X = -1) + \mathbb{P}(X = 1), & \text{if } z = 1 \\ \mathbb{P}(X = -2) + \mathbb{P}(X = 2), & \text{if } z = 4 \\ \mathbb{P}(X = -3) + \mathbb{P}(X = 3), & \text{if } z = 9 \\ 0, & \text{Otherwise} \end{cases} \\ &= \begin{cases} \frac{1}{7}, & \text{if } z = 0 \\ \frac{2}{7}, & \text{if } z = 1 \\ \frac{5}{14}, & \text{if } z = 4 \\ \frac{3}{14}, & \text{if } z = 9 \\ 0, & \text{Otherwise} \end{cases} \end{aligned}$$

The DF of Y is

$$G(z) = \mathbb{P}(Y \leq z) = \begin{cases} 0, & \text{if } z < 0 \\ \frac{1}{7}, & \text{if } 0 \leq z < 1 \\ \frac{3}{7}, & \text{if } 1 \leq z < 4 \\ \frac{11}{14}, & \text{if } 4 \leq z < 9 \\ 1, & \text{if } z \geq 9. \end{cases}$$

Example 1.2. In example 1.1, directly find the DF of $Y = X^2$ (that is find DF of Y before finding the PDF of Y). Hence find the PMF of Y .

Solution: By Theorem 1.1 Y is a discrete RV with support $T = \{0, 1, 4, 9\}$. Thus the DF of Y is

$$\begin{aligned} G(z) = \mathbb{P}(Y \leq z) = \mathbb{P}(X^2 \leq z) &= \begin{cases} 0, & \text{if } z < 0 \\ \mathbb{P}(X^2 = 0), & \text{if } 0 \leq z < 1 \\ \mathbb{P}(X^2 = 0) + \mathbb{P}(X^2 = 1), & \text{if } 1 \leq z < 4 \\ \mathbb{P}(X^2 = 0) + \mathbb{P}(X^2 = 1) + \mathbb{P}(X^2 = 4), & \text{if } 4 \leq z < 9 \\ 0, & z \geq 9 \end{cases} \\ &= \begin{cases} 0, & \text{if } z < 0 \\ \frac{1}{7}, & \text{if } 0 \leq z < 1 \\ \frac{1}{7} + \frac{1}{7} + \frac{1}{7} = \frac{3}{7}, & \text{if } 1 \leq z < 4 \\ \frac{1}{7} + \frac{1}{7} + \frac{1}{7} + \frac{3}{14} = \frac{11}{14}, & \text{if } 4 \leq z < 9 \\ 0, & z \geq 9 \end{cases} \end{aligned}$$

The PMF of Y is

$$g(z) = \begin{cases} G(z) - G(z-), & \text{if } z \in T \\ 0, & \text{Otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{7}, & \text{if } z = 0 \\ \frac{2}{7}, & \text{if } z = 1 \\ \frac{5}{14}, & \text{if } z = 4 \\ \frac{3}{14}, & \text{if } z = 9 \\ 0, & \text{Otherwise} \end{cases}$$

1.1 Probability distribution of a function of a continuous random variable

Let X be a continuous RV with DF F , PDF $f(\cdot)$ and support

$$S = \{x \in \mathbb{R} : F(x+h) - F(x-h) = \int_{x-h}^{x+h} f(t)dt > 0, \forall h > 0\}.$$

For convenience assume that $S = [a, b]$ and $\{x \in \mathbb{R} : f(x) > 0\} = (a, b)$, for some $-\infty \leq a < b \leq \infty$.

Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that h is strictly monotone and differentiable function on S . Then $Z = h(X)$ is a RV with DF

$$G(z) = \mathbb{P}(Z \leq z) = \mathbb{P}(h(X) \leq z), \quad z \in \mathbb{R}.$$

For any sets $A \subseteq \mathbb{R}$ and $B \subseteq \mathbb{R}$, define

$$h(A) = \{h(x) : x \in A\}$$

and

$$h^{-1}(B) = \{x \in \mathbb{R} : h(x) \in B\}.$$

Clearly $\mathbb{P}(X \in (a, b)) = 1$ and therefore $\mathbb{P}(h(X) \in h((a, b))) = 1$.

case I: $h(\cdot)$ is strictly increasing on S : We have

$$\mathbb{P}(h(a) < Z < h(b)) = 1.$$

Therefore, for $z < h(a)$, $\mathbb{P}(Z \leq z) = 0$ and for $z \geq h(b)$, $\mathbb{P}(Z \leq z) = 1$. For $h(a) < z < h(b)$

$$\begin{aligned} G(z) &= \mathbb{P}(h(X) \leq z) = \mathbb{P}(X \leq h^{-1}(z)) \\ &= \int_{-\infty}^{h^{-1}(z)} f(t)dt = \int_a^{h^{-1}(z)} f(t)dt \\ &= \int_{h(a)}^z f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| dy \end{aligned}$$

Thus

$$G(z) = \begin{cases} 0, & \text{if } z < h(a) \\ \int_{h(a)}^z f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| dy, & \text{if } h(a) \leq z < h(b) \\ 1, & z \geq h(b). \end{cases}$$

Since f is continuous on (a, b) it follows that $G(z)$ is differentiable everywhere except possibly at $z = h(a)$ and $z = h(b)$. Moreover,

$$G'(z) = \begin{cases} f(h^{-1}(z)) \left| \frac{dh^{-1}(z)}{dz} \right|, & \text{if } h(a) < z < h(b) \\ 0, & \text{Otherwise.} \end{cases}$$

and

$$\int_{-\infty}^{\infty} G' dz = \int_{h(a)}^{h(b)} f(h^{-1}(z)) \left| \frac{dh^{-1}(z)}{dz} \right| = \int_a^b f(t) dt = 1.$$

It follows that Z is a continuous RV with PDF

$$g(z) = \begin{cases} f(h^{-1}(z)) \left| \frac{dh^{-1}(z)}{dz} \right|, & \text{if } h(a) < z < h(b) \\ 0, & \text{Otherwise.} \end{cases}$$

and support $S = [h(a), h(b)]$.

Case II: $h(\cdot)$ is strictly decreasing on S : We have

$$\mathbb{P}(h(b) < Z < h(a)) = 1.$$

Therefore, for $z \leq h(b)$, $\mathbb{P}(Z \leq z) = 0$ and for $z \geq h(a)$, $\mathbb{P}(Z \leq z) = 1$. For $h(b) < z < h(a)$

$$\begin{aligned} G(z) &= \mathbb{P}(h(X) \leq z) = \mathbb{P}(X \geq h^{-1}(z)) \\ &= \int_{h^{-1}(z)}^{\infty} f(t) dt = \int_{h^{-1}(z)}^b f(t) dt \\ &= \int_{h(b)}^z f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| dy \end{aligned}$$

Thus

$$G(z) = \begin{cases} 0, & \text{if } z \leq h(b) \\ \int_{h(b)}^z f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| dy, & \text{if } h(b) < z < h(a) \\ 0, & \text{if } z \geq h(a). \end{cases}$$

Since f is continuous on (a, b) it follows that $G(z)$ is differentiable everywhere except possibly at $z = h(a)$ and $z = h(b)$. Moreover,

$$G'(z) = \begin{cases} f(h^{-1}(z)) \left| \frac{dh^{-1}(z)}{dz} \right|, & \text{if } h(b) < z < h(a) \\ 0, & \text{Otherwise.} \end{cases}$$

and

$$\int_{-\infty}^{\infty} G' dz = \int_{h(b)}^{h(a)} f(h^{-1}(z)) \left| \frac{dh^{-1}(z)}{dz} \right| = \int_a^b f(t) dt = 1.$$

It follows that Z is a continuous RV with PDF

$$g(z) = \begin{cases} f(h^{-1}(z)) \left| \frac{dh^{-1}(z)}{dz} \right|, & \text{if } h(b) < z < h(a) \\ 0, & \text{Otherwise.} \end{cases}$$

and support $S = [h(a), h(b)]$.

Combining Case I and II, we get the following result.

Theorem 1.2. *Let X be a continuous RV with support $S = [a, b]$, for some $-\infty \leq a < b \leq \infty$. Suppose that $\{x \in \mathbb{R} : f(x) > 0\} = (a, b)$ and that f is continuous on (a, b) . Then $Z = h(X)$ is a continuous RV with PDF*

$$\begin{cases} f(h^{-1}(z)) \left| \frac{dh^{-1}(z)}{dz} \right|, & \text{if } z \in h((a, b)) \\ 0, & \text{Otherwise.} \end{cases}$$

and support $S = [\min\{h(a), h(b)\}, \max\{h(a), h(b)\}]$.

The following theorem is a generalization of the above result and can be proved on similar lines.

Theorem 1.3. Let X be a continuous RV with support $S = \cup_{i \in \Lambda} [a_i, b_i]$, where Λ is a countable set and $[a_i, b_i]$'s $i \in \Lambda$ are disjoint intervals. Suppose that $\{x \in \mathbb{R} : f(x) > 0\} = \cup_{i \in \Lambda} (a_i, b_i)$ and that f is continuous in each (a_i, b_i) , $i \in \Lambda$. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a function that is differentiable and strictly monotone in each (a_i, b_i) , $i \in \Lambda$ (h may be monotonic increasing in some (a_i, b_i) and monotonic decreasing in some (a_i, b_i)). Let $h_i^{-1}(\cdot)$ be the inverse function of h on (a_i, b_i) , $i \in \Lambda$. Then $Z = h(X)$ is a continuous RV with PDF

$$g(z) = \sum_{j \in \Lambda} f(h_j^{-1}(z)) \left| \frac{d}{dz} h_j^{-1}(z) \right| \mathbf{I}_{(h(a_j, b_j))}(z).$$

Remark 1.1. Theorem 1.1 hold even in situations where the function h is differentiable everywhere except possibly at a finite number of points in S .

Example 1.3. Let X be a RV with PDF

$$f(x) = \begin{cases} 3x^2, & \text{if } 0 < x < 1 \\ 0, & \text{Otherwise.} \end{cases}$$

Find the PDF and DF of $Y = \frac{1}{X^2}$. What is the support of DF of Y .

Solution: The support of F is $[0, 1]$ and $\{x \in \mathbb{R} : f(x) > 0\} = (0, 1)$. Moreover f is continuous on $(0, 1)$ and $h(x) = \frac{1}{x^2}$ is differentiable and strictly monotone on $(0, 1)$. Here $h((0, 1)) = (1, \infty)$,

$$h^{-1}(y) = \frac{1}{\sqrt{y}}$$

and

$$\frac{d}{dy} h^{-1}(y) = -\frac{1}{2y\sqrt{y}}$$

where $y \in (1, \infty)$. Thus $Y = X^2$ is a continuous RV with PDF

$$\begin{aligned} g(y) &= f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| \mathbf{I}_{h((0,1))}(y) \\ &= f(h^{-1}(y)) \left| \frac{dh^{-1}(y)}{dy} \right| \mathbf{I}_{(1,\infty)}(y) \\ &= \begin{cases} \frac{3}{y} \times \frac{1}{2y\sqrt{y}}, & \text{if } y > 1 \\ 0, & \text{Otherwise.} \end{cases} \\ &= \begin{cases} \frac{3}{2y^2\sqrt{y}}, & \text{if } y > 1 \\ 0, & \text{Otherwise.} \end{cases} \end{aligned}$$

The DF of Y is

$$\begin{aligned} G(y) &= \int_{-\infty}^y g(t) dt \\ &= \begin{cases} 0, & \text{if } y < 1 \\ \int_1^y \frac{3}{2t^2\sqrt{t}} dt, & \text{if } y > 1 \end{cases} = \begin{cases} 0, & \text{if } y < 1 \\ 1 - \frac{1}{y^{3/2}}, & \text{if } y > 1. \end{cases} \end{aligned}$$

Clearly the support of G is $[1, \infty)$.

Example 1.4. Let X be RV with PDF

$$f(x) = \begin{cases} \frac{|x|}{2}, & \text{if } -1 < x < 1 \\ \frac{x}{3}, & \text{if } 1 \leq x \leq 2 \\ 0, & \text{Otherwise} \end{cases}$$

and let $Y = X^2$.

(a) Find the PDF of Y directly and hence find the DF of Y .

(b) Find the DF of Y and hence find the PDF of Y .

(c) Find the support of DF of Y .

Solution:

(a) We apply the transformation theorem for piecewise monotone functions.

The support of X is

$$S = (-1, 1) \cup [1, 2].$$

Partition S into intervals

$$(-1, 0), \quad (0, 1), \quad (1, 2),$$

on each of which $h(x) = x^2$ is strictly monotone. Hence the theorem applies.

For $y > 0$, the inverse functions of $h(x) = x^2$ on these intervals are

$$h_1^{-1}(y) = -\sqrt{y}, \quad h_2^{-1}(y) = \sqrt{y}, \quad h_3^{-1}(y) = \sqrt{y},$$

with

$$\left| \frac{d}{dy} h_i^{-1}(y) \right| = \frac{1}{2\sqrt{y}}.$$

PDF of Y :

Case $0 < y < 1$: Both $x = -\sqrt{y} \in (-1, 0)$ and $x = \sqrt{y} \in (0, 1)$ contribute.

$$f(-\sqrt{y}) = \frac{\sqrt{y}}{2}, \quad f(\sqrt{y}) = \frac{\sqrt{y}}{2}.$$

Thus,

$$\begin{aligned} g_Y(y) &= [f(-\sqrt{y}) + f(\sqrt{y})] \frac{1}{2\sqrt{y}} \\ &= \frac{1}{2}, \quad 0 < y < 1. \end{aligned}$$

Case $1 < y < 4$: Only $x = \sqrt{y} \in (1, 2)$ contributes.

$$f(\sqrt{y}) = \frac{\sqrt{y}}{3},$$

and hence

$$g_Y(y) = \frac{\sqrt{y}}{3} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{6}, \quad 1 < y < 4.$$

Therefore, the pdf of Y is

$$g_Y(y) = \begin{cases} \frac{1}{2}, & 0 < y < 1, \\ \frac{1}{6}, & 1 < y < 4, \\ 0, & \text{otherwise.} \end{cases}$$

Distribution function of Y :

For $y \leq 0$, $F_Y(y) = 0$. For $0 < y < 1$,

$$F_Y(y) = \int_0^y \frac{1}{2} dt = \frac{y}{2}.$$

For $1 \leq y < 4$,

$$F_Y(y) = \int_0^1 \frac{1}{2} dt + \int_1^y \frac{1}{6} dt = \frac{1}{2} + \frac{y-1}{6} = \frac{y+2}{6}.$$

For $y \geq 4$, $F_Y(y) = 1$.

Hence,

$$F_Y(y) = \begin{cases} 0, & y \leq 0, \\ \frac{y}{2}, & 0 < y < 1, \\ \frac{y+2}{6}, & 1 \leq y < 4, \\ 1, & y \geq 4. \end{cases}$$

(b) Finding the CDF first

For $y \geq 0$,

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(-\sqrt{y} \leq X \leq \sqrt{y}).$$

Case 1: $0 \leq y < 1$

$$F_Y(y) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{|x|}{2} dx = 2 \int_0^{\sqrt{y}} \frac{x}{2} dx = \frac{y}{2}.$$

Case 2: $1 \leq y < 4$

$$F_Y(y) = \int_{-1}^1 \frac{|x|}{2} dx + \int_1^{\sqrt{y}} \frac{x}{3} dx = \frac{1}{2} + \frac{y-1}{6} = \frac{y+2}{6}.$$

For $y \geq 4$, $F_Y(y) = 1$. Differentiating $F_Y(y)$ yields the same pdf obtained in part (a).

(c) Support of Y

Since $X \in (-1, 1) \cup [1, 2]$, we have

$$Y = X^2 \in [0, 4].$$

Hence, the support of Y is $\boxed{[0, 4]}$.

2 Expectation (or mean) of random variables

Let X be a discrete RV with PMF $f(\cdot)$ and support S . For any $x \in S$, $f(x)$ gives an idea about proportion of times we will observe the event $\{X = x\}$ if the experiment is repeated a large number of times. Thus $\sum_{x \in S} xf(x)$ represents the mean (or expectation) value of RV X if the experiment is repeated a large number of times.

Similarly if X is a continuous RV with PDF $f(\cdot)$ then

$$\int_{-\infty}^{\infty} xf(x)dx \quad (\text{provided the integral is finite})$$

represents the mean value of RV X .

Definition 2.1. (a) Let X be a discrete RV with PMF $f(\cdot)$ and support S . We say that the expected value of X which we denote by $\mathbb{E}(X)$ is finite and equals

$$\mathbb{E}(X) = \sum_{x \in S} xf(x),$$

provided $\sum_{x \in S} |x|f(x) < \infty$.

(b) Let X be a continuous RV with PDF $f(\cdot)$ and support S . We say that the expected value of X is finite and equals

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x)dx,$$

provided $\int_{-\infty}^{\infty} |x|f(x)dx < \infty$.

Example 2.1. (a) Let X be a discrete RV with PMF

$$f(x) = \begin{cases} \left(\frac{1}{2}\right)^x, & x \in \{1, 2, 3, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$

Show that $\mathbb{E}(X)$ is finite. Find $\mathbb{E}(X)$.

(b) Let X be a RV with PMF

$$f(x) = \begin{cases} \frac{3}{\pi^2 x^2}, & x \in \{\pm 1, \pm 2, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$

Show that $\mathbb{E}(X)$ is not finite.

(c) Let X be a continuous RV with PDF

$$f(x) = \frac{e^{-|x|}}{2}, \quad -\infty < x < \infty$$

Show that $\mathbb{E}(X)$ is finite. Find $\mathbb{E}(X)$.

(d) Let X be a continuous RV with PDF

$$f(x) = \frac{1}{\pi} \cdot \frac{1}{1+x^2}, \quad -\infty < x < \infty$$

Show that $\mathbb{E}(X)$ is not finite.

Solution:

(a) The expectation of a discrete random variable is

$$\mathbb{E}(X) = \sum_{x=1}^{\infty} xf(x) = \sum_{x=1}^{\infty} x \left(\frac{1}{2}\right)^x.$$

Using the standard identity

$$\sum_{x=1}^{\infty} xr^x = \frac{r}{(1-r)^2}, \quad 0 < r < 1,$$

with $r = \frac{1}{2}$, we obtain

$$\mathbb{E}(X) = \frac{\frac{1}{2}}{(1-\frac{1}{2})^2} = 2.$$

Hence $\mathbb{E}(X)$ is finite and $\mathbb{E}(X) = 2$.

(b) Since the PMF is symmetric about 0, we examine

$$\mathbb{E}(|X|) = \sum_{x=\pm 1, \pm 2, \dots} |x|f(x) = 2 \sum_{x=1}^{\infty} x \frac{3}{\pi^2 x^2} = \frac{6}{\pi^2} \sum_{x=1}^{\infty} \frac{1}{x}.$$

The harmonic series diverges, hence $\mathbb{E}(|X|) = \infty$. Therefore $\mathbb{E}(X)$ is not finite.

(c) The expectation is

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} xf(x) dx = \frac{1}{2} \int_{-\infty}^{\infty} xe^{-|x|} dx.$$

Since $f(x)$ is symmetric about 0 and x is an odd function, the integrand is odd. Hence

$$\mathbb{E}(X) = 0.$$

Moreover,

$$\int_{-\infty}^{\infty} |x|f(x) dx < \infty,$$

so $\mathbb{E}(X)$ is finite.

(d) The expectation (if it exists) is

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x \frac{1}{\pi(1+x^2)} dx.$$

Although the integrand is odd, we must check absolute integrability:

$$\int_{-\infty}^{\infty} |x| \frac{1}{\pi(1+x^2)} dx = \frac{2}{\pi} \int_0^{\infty} \frac{x}{1+x^2} dx.$$

Since

$$\int_0^{\infty} \frac{x}{1+x^2} dx = \frac{1}{2} \int_0^{\infty} \frac{1}{u} du = \infty,$$

the integral diverges. Hence $\mathbb{E}(X)$ is not finite.

Example 2.2. To make some money a gambler plays a sequence of fair games with the following strategy. In the first, bet Rs. 1 million. If the first bet is lost he doubles his bet in the second game. He keeps on doubling his bet until he wins a game. If he wins in the k th game then

$$\text{Investment} = 1 + 2 + 4 + \dots + 2^{K-1} = 2^K - 1 \text{ million rupees}$$

$$\text{Win} = 2^K \text{ million rupees}$$

Total earning if he wins on k th game = 1 million rupee. The above scheme seems to be fool proof for earning Rs. 1 million. By this logic all gamblers should be billionaires!

X : the amount of money bet on the last game (the game he wins). Then

$$\mathbb{P}(X = 2^k - 1) = \frac{1}{2^k}, \quad k = 1, 2, \dots$$

$$\mathbb{E}(X) = \sum_{k=1}^{\infty} 2^{k-1} \times \frac{1}{2^k} = \infty.$$

Thus enormous amount of money should be required.

Theorem 2.1. Let X be a discrete or continuous RV. Then

$$\mathbb{E}(X) = \int_0^{\infty} \mathbb{P}(X > y) dy - \int_{-\infty}^0 \mathbb{P}(X < y) dy,$$

provided $\mathbb{E}(X)$ is finite.

Proof. We shall provide the proof for the case when X is a continuous RV with PDF say f . we have

$$\begin{aligned}
\mathbb{E}(X) &= \int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^0 xf(x)dx + \int_0^{\infty} xf(x)dx \\
&= - \int_{-\infty}^0 \int_x^0 f(x)dydx + \int_0^{\infty} \int_0^x f(x)dx \\
&= - \int_{-\infty}^0 \int_{-\infty}^y f(x)dx dy + \int_0^{\infty} \int_y^{\infty} f(x)dx dy \\
&= - \int_{-\infty}^0 \mathbb{P}(X < y)dy + \int_0^{\infty} \mathbb{P}(X > y)dy.
\end{aligned}$$

□

Corollary 2.1.1. (a) Suppose that X is a discrete continuous RV with $\mathbb{P}(X \geq 0) = 1$. Then

$$\mathbb{E}(X) = \int_0^{\infty} \mathbb{P}(X > y)dy.$$

(b) Suppose that $\mathbb{P}(X \in \{0, \pm 1, \pm 2, \dots\}) = 1$. Then

$$\mathbb{E}(X) = \sum_{n=1}^{\infty} \mathbb{P}(X \geq n) - \mathbb{P}(X \leq -n).$$

(c) Suppose that $\mathbb{P}(X \in \{0, 1, 2, \dots\}) = 1$. Then

$$\mathbb{E}(X) = \sum_{n=1}^{\infty} \mathbb{P}(X \geq n).$$

Proof. Exercise. □

The following theorem suggest that for any RV X and any function $h : \mathbb{R} \rightarrow \mathbb{R}$, $\mathbb{E}(h(X))$ can be directly found using PMF/PDF of X .

Theorem 2.2. (a) Let X be a discrete RV with support S and PMF f . Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a given function and let $Z = h(X)$. Then

$$\mathbb{E}(Z) = \sum_{x \in S} h(x)f(x)$$

provided $\sum_{x \in S} |h(x)|f(x) < \infty$.

(b) Let X be a continuous RV with support S and PDF f . Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be a given function and let $T = h(X)$. Then

$$\mathbb{E}(Z) = \int_{-\infty}^{\infty} h(x)f(x)$$

provided $\int_{-\infty}^{\infty} |h(x)|f(x) < \infty$.

Proof. We shall provide the proof for (a). The proof of (b) follows on similar lines. The support of $T = h(X)$ is $T = h(S)$.

$$\begin{aligned}
\mathbb{E}(Z) &= \sum_{t \in T} t\mathbb{P}(Z = t) = \sum_{t \in T} t\mathbb{P}(h(X) = t) = \sum_{t \in T} t \left\{ \sum_{x \in S: h(x)=t} \mathbb{P}(X = x) \right\} \\
&= \sum_{\{x \in S: h(x)=t\}} \sum_{t \in T} t\mathbb{P}(X = x) = \sum_{t \in T} \sum_{\{x \in S: h(x)=t\}} h(x)\mathbb{P}(X = x) \\
&= \sum_{x \in S} h(x)\mathbb{P}(X = x).
\end{aligned}$$

□

Example 2.3. (a) Let the RV have the PMF

$$f(x) = \begin{cases} \frac{1}{6}, & \text{if } x = -2, -1, 0, 1, 2, 3 \\ 0, & \text{otherwise.} \end{cases}$$

Find $\mathbb{E}(X^2)$.

(b) Let the RV X have the PDF

$$f(x) = \begin{cases} 2x, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find $\mathbb{E}(X^3)$.

Solution:

(a)

$$\mathbb{E}(X^2) = \sum_{x \in S} x^2 f(x) = 4 \times \frac{1}{6} + 1 \times \frac{1}{6} + 0 \times \frac{1}{6} + 1 \times \frac{1}{6} + 4 \times \frac{1}{6} + 9 \times \frac{1}{6} = \frac{19}{6}$$

(b)

$$\mathbb{E}(X^3) = \int_{-\infty}^{\infty} x^3 f(x) dx = 2 \int_0^1 x^4 dx = \frac{2}{5}.$$

Theorem 2.3. Let X be a discrete or continuous RV with PMF/PDF f and $h_i : \mathbb{R} \rightarrow \mathbb{R}$ ($i = 1, 2, \dots, n$) be given functions.

(a) Then, for real constants $c_1, \dots, c_n$

$$\mathbb{E}\left(\sum_{i=1}^n c_i h_i(X)\right) = \sum_{i=1}^n c_i \mathbb{E}(h_i(X)),$$

provided involved expectations are finite.

(b) Let $h_1(x) \leq h_2(x)$, $\forall x \in S$. Then

$$\mathbb{E}(h_1(x)) \leq \mathbb{E}(h_2(x))$$

provided involved expectations are finite. In particular if $\mathbb{E}(X)$ is finite and $\mathbb{P}(a \leq X \leq b) = 1$, for some real constant a and b ($a < b$) then $a \leq \mathbb{E}(X) \leq b$.

(c) If $\mathbb{P}(X \geq 0) = 1$ and $\mathbb{E}(X) = 0$, then $\mathbb{P}(X = 0) = 1$.

(d) If $\mathbb{E}(X)$ is finite then $|\mathbb{E}(X)| \leq \mathbb{E}(|X|)$.

(e) For real constants a and b $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$ provided involved expectations are finite.

Proof. The proofs for assertion (a), (b), and (e) are obvious.

(c) We will provide the proof for the case when X is a discrete RV. Suppose X is a discrete random variable with PMF $p(x)$ and $\mathbb{P}(X \geq 0) = 1$, $\mathbb{E}(X) = 0$.

First observe that

$$\{X > 0\} = \bigcup_{n=1}^{\infty} \left\{X \geq \frac{1}{n}\right\}.$$

Since $\{X \geq \frac{1}{n}\} \uparrow \{X > 0\}$, by continuity of probability from below,

$$\mathbb{P}(X > 0) = \lim_{n \rightarrow \infty} \mathbb{P}\left(X \geq \frac{1}{n}\right).$$

Now,

$$\mathbb{P}\left(X \geq \frac{1}{n}\right) = \sum_{x \geq 1/n} p(x).$$

For $x \geq \frac{1}{n}$, we have $nx \geq 1$. Hence

$$p(x) \leq nx p(x).$$

Summing over all $x \geq \frac{1}{n}$,

$$\sum_{x \geq 1/n} p(x) \leq \sum_{x \geq 1/n} nx p(x) = n \sum_{x \geq 1/n} x p(x) \leq n \sum_x x p(x).$$

Thus,

$$\mathbb{P}\left(X \geq \frac{1}{n}\right) \leq n\mathbb{E}(X).$$

Since $\mathbb{E}(X) = 0$, we get

$$\mathbb{P}\left(X \geq \frac{1}{n}\right) = 0 \quad \text{for all } n.$$

Therefore,

$$\mathbb{P}(X > 0) = 0.$$

Because $\mathbb{P}(X \geq 0) = 1$, it follows that

$$\mathbb{P}(X = 0) = 1.$$

(d) We have

$$\begin{aligned} -|X| &\leq X \leq |X| \\ \implies \mathbb{E}(-|X|) &\leq \mathbb{E}(X) \leq \mathbb{E}(|X|) \\ \implies |\mathbb{E}(X)| &\leq \mathbb{E}(|X|). \end{aligned}$$

□

Some special expectations

- (i) $h(x) = x$; $\mathbb{E}(X) = \mu$ = mean of (distribution of) X .
- (ii) $h(x) = x^r$, $r \in \{1, 2, \dots\}$; $\mathbb{E}(X^r) = \mu_r$ = r th moment of X about origin.
- (iii) $h(x) = |x|^r$, $r \in \{1, 2, \dots\}$; $\mathbb{E}(|X|^r) = r$ th absolute moment of X about origin.
- (iv) $h(x) = (x - \mu)^r$, $r \in \{1, 2, \dots\}$; $\mathbb{E}((X - \mu)^r) = \mu_r$ = r th moment of X about its mean or r -th central moment.
- (v) $\mathbb{E}((X - \mu)^2) = \sigma^2$ = variance of X (also denoted by $\text{Var}(X)$). σ = standard deviation of X .

Remark 2.1. (a)

$$\begin{aligned} \text{Var}(X) = \sigma^2 &= \mathbb{E}((X - \mu)^2) = \mathbb{E}(X^2 - 2\mu X + (\mu)^2) \\ &= \mathbb{E}(X^2) - 2\mu\mathbb{E}(X) + \mu^2 = \mathbb{E}(X^2) - 2\mu^2 + \mu^2 \\ &= \mathbb{E}(X^2) - \mu^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2. \end{aligned}$$

(b) Since $(X - \mu)^2 \geq 0$, we have

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}((X - \mu)^2) \geq 0 \\ \implies \mathbb{E}(X^2) &\geq (\mathbb{E}(X))^2. \end{aligned}$$

(c)

$$\begin{aligned}\text{Var}(X) = 0 &\implies \mathbb{E}((X - \mu)^2) = 0 \\ &\implies \mathbb{P}(X = \mathbb{E}(X)) = 1.\end{aligned}$$

Theorem 2.4. Let X be a RV such that $\mathbb{E}(|X|^s) < \infty$, for some $s > 0$. Then $\mathbb{E}(|X|^r) < \infty$, $\forall 0 < r < s$.

Proof. Note that

$$\begin{aligned}|X|^r &\leq \max\{|X|^s, 1\} \leq |X|^s + 1 \\ \implies \mathbb{E}(|X|^r) &\leq \mathbb{E}(|X|^s + 1) = \mathbb{E}(|X|^s) + 1 < \infty.\end{aligned}$$

Thus the result follows. □

3 Moment generating function (MGF)

Let X be a RV with DF F and PMF/PDF f .

Definition 3.1. We say that the moment generating function of X (denoted by $M_X(t)$) exists and equals

$$M_X(t) = \mathbb{E}(e^{tX})$$

provided $\mathbb{E}(e^{tX})$ is finite in $(-h, h)$ for some $h > 0$.

Theorem 3.1. Let X be a RV with MGF M_X that is finite on $(-h, h)$, $h > 0$. Then

- (a) for each $r \in \{1, 2, \dots\}$, $\mu_r = \mathbb{E}(X^r)$ is finite;
- (b) for each $r \in \{1, 2, \dots\}$, $\mu_r = \mathbb{E}(X^r) = M_X^{(r)}(0)$, where

$$M_X^{(r)}(0) = \left[\frac{d^r}{dt^r} M_X(t) \right]_{t=0},$$

the r th derivative of $M_X(t)$ at $t = 0$;

- (c) $M_X(t) = \sum_{r=0}^{\infty} \frac{t^r}{(r-1)!} \mu_r$, $t \in (-h, h)$ so that μ_r is equal to coefficient of $\frac{t^r}{(r-1)!}$ ($r = 1, 2, \dots$) in the Maclaurin's series expansion of $M_X(t)$ around $t = 0$.

Proof. (a) We have

$$\begin{aligned}\mathbb{E}(e^{tX}) &< \infty, \forall t \in (-h, h) \\ \implies \int_{-\infty}^0 e^{tx} f(x) dx &< \infty, \forall t \in (-h, h) \text{ and } \int_0^{\infty} e^{tx} f(x) dx < \infty, \forall t \in (-h, h) \\ \implies \int_{-\infty}^0 e^{-t|x|} f(x) dx &< \infty, \forall t \in (-h, h) \text{ and } \int_0^{\infty} e^{t|x|} f(x) dx < \infty, \forall t \in (-h, h) \\ \implies \int_{-\infty}^0 e^{|t||x|} f(x) dx &< \infty, \forall t \in (-h, h) \text{ and } \int_0^{\infty} e^{|t||x|} f(x) dx < \infty, \forall t \in (-h, h) \\ \implies \int_{-\infty}^{\infty} e^{|t|x|} f(x) dx &< \infty, \forall t \in (-h, h)\end{aligned}$$

here $f(\cdot)$ denotes the PDF of RV X .

Fix $r \in \{1, 2, \dots\}$ and $t \in (-h, h) - \{0\}$. Then $\lim_{|x| \rightarrow \infty} \frac{|x|^r}{e^{|tx|}} = 0$ and therefore there exists a positive real

number $A_{r,t}$ such that $|x|^r < e^{|tx|}$, $\forall |x| > A_{r,t}$. Therefore,

$$\begin{aligned}
\mathbb{E}(|X|^r) &= \int_{-\infty}^{\infty} |x|^r f(x) dx \\
&= \int_{|x| \leq A_{r,t}} |x|^r f(x) dx + \int_{|x| \geq A_{r,t}} |x|^r f(x) dx \\
&\leq A_{r,t}^r \int_{|x| \leq A_{r,t}} f(x) dx + \int_{|x| \geq A_{r,t}} e^{|tx|} f(x) dx \\
&\leq A_{r,t}^r + \int_{-\infty}^{\infty} e^{|tx|} f(x) dx < \infty, \quad \forall r = 1, 2, \dots
\end{aligned}$$

(b)

$$\begin{aligned}
M_X(t) &= \int_{-\infty}^{\infty} e^{tx} f(x) dx \\
M_X^{(r)}(t) &= \frac{d^r}{dt^r} \int_{-\infty}^{\infty} e^{tx} f(x) dx
\end{aligned}$$

Using the arguments of advanced calculus it can be shown that if $M_X(t) = \mathbb{E}(e^{tX}) < \infty$, $\forall t \in (-h, h)$ then the derivative can be passed through the integral sign. Therefore,

$$\begin{aligned}
M_X^{(r)}(t) &= \int_{-\infty}^{\infty} \frac{d^r}{dt^r} (e^{tx} f(x)) dx \\
&= \int_{-\infty}^{\infty} x^r e^{tx} f(x) dx, \quad r = 1, 2, \dots \\
M_X^{(r)}(0) &= \int_{-\infty}^{\infty} x^r f(x) dx = \mathbb{E}(X^r).
\end{aligned}$$

(c)

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f(x) dx = \int_{-\infty}^{\infty} \left(\sum_{r=0}^{\infty} \frac{t^r x^r}{(r!)} \right) f(x) dx.$$

Under the assumption that $M_X(t) = \mathbb{E}(e^{tX}) < \infty$, $\forall t \in (-h, h)$, using the arguments of advanced calculus, it can be shown that the summation sign can be passed through the integral sign. Thus

$$M_X(t) = \sum_{r=0}^{\infty} \frac{t^r}{(r!)} \int_{-\infty}^{\infty} x^r f(x) dx = \sum_{r=0}^{\infty} \frac{t^r}{(r!)} \mathbb{E}(X^r), \quad r = 1, 2, \dots$$

□

Corollary 3.1.1. *Under the notation and assumption of the above theorem, let $\Psi_X(t) = \ln M_X(t)$, $t \in (-h, h)$. Then*

$$\mu_1 = \mathbb{E}(X) = \Psi_X^{(1)}(0)$$

and

$$\text{Var}(X) = \Psi_X^{(2)}(0)$$

Proof. For $t \in (-h, h)$,

$$\Psi_X^{(1)}(t) = \frac{M_X^{(1)}(t)}{M_X(t)}$$

and

$$\Psi_X^{(2)}(t) = \frac{M_X(t)M_X^{(2)}(t) - (M_X^{(1)}(t))^2}{(M_X(t))^2}$$

$$\implies \Psi_X^{(1)}(0) = M_X^{(1)}(0) = \mathbb{E}(X) \quad (M_X(0) = 1)$$

and

$$\Psi_X^{(2)}(0) = M_X^{(2)}(0) - (M_X^{(1)}(0))^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = \text{Var}(X).$$

□

Example 3.1. (a) Let X be a discrete RV with PMF

$$f_X(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & \text{if } x = 0, 1, 2, \dots \\ 0, & \text{otherwise.} \end{cases}$$

where $\lambda > 0$. Show that the MGF of X exists and is finite on whole $\mathbb{R}$. Find $M_X(t)$, mean, and variance of X and $\mathbb{E}(X^3)$.

(b) Let X be a continuous RV with PDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x > 0 \dots \\ 0, & \text{otherwise.} \end{cases}$$

where $\lambda > 0$. Find the MGF of X , mean and variance of X and $\mathbb{E}(X^r)$, $r = 1, 2, \dots$, provided they exist.

(c) Let X be a continuous RV having the PDF called Cauchy PDF. Show that the MGF of X does not exist.

Solution:

(a) Let

$$f_X(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!}, & x = 0, 1, 2, \dots, \\ 0, & \text{otherwise,} \end{cases} \quad \lambda > 0.$$

Then $X \sim \text{Poisson}(\lambda)$.

1. Existence of the MGF

$$\begin{aligned} M_X(t) &= \mathbb{E}(e^{tX}) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda e^t)^x}{x!}. \end{aligned}$$

Using

$$\sum_{x=0}^{\infty} \frac{a^x}{x!} = e^a,$$

we obtain

$$M_X(t) = e^{-\lambda} e^{\lambda e^t} = \exp(\lambda(e^t - 1)).$$

Since this expression is finite for all $t \in \mathbb{R}$, the MGF exists on the whole $\mathbb{R}$.

2. MGF

$M_X(t) = \exp(\lambda(e^t - 1))$

3. Moments using power series expansion

Recall

$$M_X(t) = \sum_{n=0}^{\infty} \frac{\mathbb{E}(X^n)}{n!} t^n.$$

First expand:

$$e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots$$

Hence,

$$\lambda(e^t - 1) = \lambda \left(t + \frac{t^2}{2} + \frac{t^3}{6} + \cdots \right).$$

Let

$$A(t) = \lambda t + \frac{\lambda}{2} t^2 + \frac{\lambda}{6} t^3 + O(t^4).$$

Then

$$M_X(t) = e^{A(t)}.$$

Using

$$e^A = 1 + A + \frac{A^2}{2} + \frac{A^3}{6} + O(t^4),$$

we compute:

$$A^2 = \lambda^2 t^2 + \lambda^2 t^3 + O(t^4),$$

$$\frac{A^2}{2} = \frac{\lambda^2}{2} t^2 + \frac{\lambda^2}{2} t^3,$$

$$\frac{A^3}{6} = \frac{\lambda^3}{6} t^3.$$

Thus,

$$M_X(t) = 1 + \lambda t + \frac{\lambda + \lambda^2}{2} t^2 + \frac{\lambda + 3\lambda^2 + \lambda^3}{6} t^3 + O(t^4).$$

Comparing with

$$M_X(t) = 1 + \mathbb{E}(X)t + \frac{\mathbb{E}(X^2)}{2} t^2 + \frac{\mathbb{E}(X^3)}{6} t^3 + \cdots,$$

we obtain:

$$\boxed{\mathbb{E}(X) = \lambda}$$

$$\boxed{\mathbb{E}(X^2) = \lambda^2 + \lambda}$$

$$\boxed{\text{Var}(X) = \lambda}$$

$$\boxed{\mathbb{E}(X^3) = \lambda^3 + 3\lambda^2 + \lambda}$$

(b) Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \\ 0, & \text{otherwise,} \end{cases}$$

where $\lambda > 0$.

Moment Generating Function

By definition,

$$M_X(t) = \mathbb{E}(e^{tX}) = \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx.$$

$$M_X(t) = \lambda \int_0^\infty e^{-(\lambda-t)x} dx.$$

The integral converges if and only if $t < \lambda$. For $t < \lambda$,

$$M_X(t) = \lambda \cdot \frac{1}{\lambda - t} = \frac{\lambda}{\lambda - t}.$$

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

Mean

$$M_X(t) = \frac{\lambda}{\lambda - t}$$

$$M'_X(t) = \frac{\lambda}{(\lambda - t)^2}$$

$$\mathbb{E}(X) = M'_X(0) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

$$\mathbb{E}(X) = \frac{1}{\lambda}$$

Second Moment and Variance

$$M''_X(t) = \frac{2\lambda}{(\lambda - t)^3}$$

$$\mathbb{E}(X^2) = M''_X(0) = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2}.$$

$$\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

r -th Moment

For $r = 1, 2, \dots$,

$$\mathbb{E}(X^r) = \int_0^\infty x^r \lambda e^{-\lambda x} dx.$$

Using the Gamma integral,

$$\int_0^\infty x^r e^{-\lambda x} dx = \frac{\Gamma(r+1)}{\lambda^{r+1}} = \frac{r!}{\lambda^{r+1}}.$$

Thus,

$$\mathbb{E}(X^r) = \lambda \cdot \frac{r!}{\lambda^{r+1}} = \frac{r!}{\lambda^r}.$$

$\mathbb{E}(X^r) = \frac{r!}{\lambda^r}, \quad r = 1, 2, \dots$

(c) Since $\mathbb{E}(X)$ is not finite, the MGF of X does not exist.

Definition 3.2. Let X and Y be two RVs with DFs F_X and F_Y , respectively. We say that X and Y have the same distribution (written as $X \stackrel{d}{=} Y$) if $F_X(x) = F_Y(x)$, $\forall x \in \mathbb{R}$.

Remark 3.1. (a) Let X and Y be discrete RVs with PMFs f_X and f_Y , respectively. Then

$$X \stackrel{d}{=} Y \iff f_X(x) = f_Y(y), \forall x \in \mathbb{R}.$$

(b) Let X and Y be continuous RVs then $X \stackrel{d}{=} Y$ iff there exist versions of PDFs f_X and f_Y of X and Y , respectively, such that $f_X(x) = f_Y(x)$, $\forall x \in \mathbb{R}$.

(c) Suppose that $X \stackrel{d}{=} Y$. Then for any function $h : \mathbb{R} \rightarrow \mathbb{R}$ $h(X) \stackrel{d}{=} h(Y)$ and hence $\mathbb{E}(h(X)) = \mathbb{E}(h(Y))$.

Theorem 3.2. Let X and Y be RVs such that for some $c > 0$, $M_X(t) = M_Y(t)$, $\forall t \in (-c, c)$. Then $X \stackrel{d}{=} Y$.

Proof. (Special case): Suppose that X and Y are discrete RVs with support $S_X = S_Y = \{1, 2, \dots\}$,

$$p_k = \mathbb{P}(X = k) \text{ and } q_k = \mathbb{P}(Y = k), k = 1, 2, \dots$$

Then

$$\begin{aligned} M_X(t) &= M_Y(t), \forall t \in (-c, c), \text{ for some } c > 0 \\ \implies \sum_{k=1}^{\infty} e^{kt} p_k &= \sum_{k=1}^{\infty} e^{kt} q_k, \forall t \in (-c, c) \\ \implies \sum_{k=1}^{\infty} s^k e^{kt} p_k &= \sum_{k=1}^{\infty} s^k e^{kt} q_k, \forall s \in (-e^{-c}, e^c) \\ \implies p_k &= q_k, \forall k = 1, 2, \dots \end{aligned}$$

Since if two power series are equal over an interval then their coefficients are also equal. Thus $X \stackrel{d}{=} Y$. $\square$

Example 3.2. For any $p \in (0, 1)$ and positive integer n , let $X_{p,n}$ be discrete RV with PMF

$$f_{p,n}(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x \in \{0, 1, 2, \dots\}, n \in \mathbb{N}, p \in (0, 1) \\ 0, & \text{otherwise.} \end{cases}$$

Such a RV (DF) is called binomial RV (DF) with n trials and probability of success p . Define $Y_{p,n} = X_{1-p,n}$. Using the MGF of $X_{p,n}$, show that $Y_{p,n} \stackrel{d}{=} X_{1-p,n}$ and $\mathbb{E}(X_{\frac{1}{2},n}) = \frac{1}{2}$.

Solution: We have

$$\begin{aligned} M_{X_{p,n}}(t) &= \mathbb{E}(e^{tX_{p,n}}) = \sum_{x=0}^n e^{tx} \binom{n}{x} p^x (1-p)^{n-x} \\ &= \sum_{x=0}^n \binom{n}{x} (e^t p)^x (1-p)^{n-x} = (1-p + pe^t)^n, t \in \mathbb{R} \end{aligned}$$

$$\begin{aligned} M_{Y_{p,n}}(t) &= \mathbb{E}(e^{tY_{p,n}}) = \mathbb{E}(e^{t(n-X_{p,n})}) = e^{nt} M_{X_{p,n}}(-t) \\ &= e^{nt} (1-p + pe^{-t})^n = (p + (1-p)e^t)^n \\ &= (1 + (1-p) + (1-p)e^t)^n = M_{X_{1-p,n}}(t), t \in \mathbb{R}. \end{aligned}$$

Thus $Y_{p,n} \stackrel{d}{=} X_{1-p,n}$.

$$\begin{aligned}
f_{Y_{p,n}} &= \mathbb{P}(Y_{p,n} = y) = \mathbb{P}(X_{p,n} = n - y) \\
&= \begin{cases} \binom{n}{n-y} p^{n-y} (1-p)^{n-(n-y)}, & n - y \in \{0, 1, 2, \dots\}, \quad n \in \mathbb{N}, \quad p \in (0, 1) \\ 0, & \text{otherwise.} \end{cases} \\
&= \begin{cases} \binom{n}{y} (1 - (1-p))^{n-y} (1-p)^y, & y \in \{0, 1, 2, \dots\}, \quad n \in \mathbb{N}, \quad p \in (0, 1) \\ 0, & \text{otherwise.} \end{cases} \\
&= f_{X_{1-p,n}}(y), \quad \forall y \in \mathbb{R}.
\end{aligned}$$

$$\implies Y_{p,n} \stackrel{d}{=} X_{1-p,n}.$$

$$\text{For } p = \frac{1}{2}, \quad X_{\frac{1}{2},n} \stackrel{d}{=} n - X_{\frac{1}{2},n},$$

$$\begin{aligned}
\implies \mathbb{E}(X_{\frac{1}{2},n}) &= \mathbb{E}(n - X_{\frac{1}{2},n}) \\
\implies \mathbb{E}(X_{\frac{1}{2},n}) &= \frac{1}{2}.
\end{aligned}$$

Example 3.3. Let X be a RV with PDF

$$f_X(t) = \frac{e^{-|x|}}{2}, \quad -\infty < x < \infty,$$

and let $Y = -X$. Show that $Y \stackrel{d}{=} X$ and show that $\mathbb{E}(X) = 0$.

1. Showing $Y \stackrel{d}{=} X$ (direct method)

For $Y = -X$, the PDF of Y is

$$f_Y(y) = f_X(-y) \left| \frac{d(-y)}{dy} \right| = f_X(-y) = \frac{1}{2} e^{-|-y|} = \frac{1}{2} e^{-|y|}.$$

Thus $f_Y(y) = f_X(y)$ for all $y \in \mathbb{R}$, and hence

$$Y \stackrel{d}{=} X.$$

2. Showing $Y \stackrel{d}{=} X$ using the MGF

The moment generating function of X is

$$M_X(t) = \mathbb{E}(e^{tX}) = \frac{1}{2} \left(\int_{-\infty}^0 e^{(t+1)x} dx + \int_0^{\infty} e^{(t-1)x} dx \right),$$

which converges for $|t| < 1$. Evaluating,

$$\int_{-\infty}^0 e^{(t+1)x} dx = \frac{1}{t+1}, \quad \int_0^{\infty} e^{(t-1)x} dx = \frac{1}{1-t}.$$

Hence

$$M_X(t) = \frac{1}{2} \left(\frac{1}{1+t} + \frac{1}{1-t} \right) = \frac{1}{1-t^2}, \quad |t| < 1.$$

For $Y = -X$,

$$M_Y(t) = \mathbb{E}(e^{tY}) = \mathbb{E}(e^{-tX}) = M_X(-t) = \frac{1}{1-t^2}.$$

Thus $M_Y(t) = M_X(t)$ for $|t| < 1$, and therefore

$$Y \stackrel{d}{=} X.$$

3. Expectation of X

The expectation of X is

$$\begin{aligned}\mathbb{E}(X) &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \frac{1}{2} \left(\int_{-\infty}^0 x e^x dx + \int_0^{\infty} x e^{-x} dx \right).\end{aligned}$$

Evaluating the integrals,

$$\int_{-\infty}^0 x e^x dx = -1, \quad \int_0^{\infty} x e^{-x} dx = 1.$$

Therefore,

$$\mathbb{E}(X) = \frac{1}{2}(-1 + 1) = 0.$$

4 Inequalities

Inequalities provide estimates of probabilities when they can not be evaluated precisely.

Theorem 4.1. Let X be a RV and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a non-negative function such that $\mathbb{E}(g(X))$ is finite. Then for any $c > 0$,

$$\mathbb{P}(g(X) \geq c) \leq \frac{\mathbb{E}(g(X))}{c}.$$

Proof. (For the case when X is continuous RV) Let $A = \{x \in \mathbb{R} : g(x) \geq c\}$ and let $f_x(\cdot)$ denote the PDF of X . Then

$$\begin{aligned}\mathbb{E}(g(X)) &= \int_{-\infty}^{\infty} g(x) f_X(x) dx = \int_{-\infty}^{\infty} g(x) [I_A(x) + I_{A^c}(x)] f_X(x) dx \\ &= \int_{-\infty}^{\infty} g(x) [I_A(x)] f_X(x) dx + \int_{-\infty}^{\infty} g(x) [I_{A^c}(x)] f_X(x) dx \\ &\geq \int_{-\infty}^{\infty} g(x) [I_A(x)] f_X(x) dx \geq c \int_{-\infty}^{\infty} [I_A(x)] f_X(x) dx \\ &= c \int_A f_X(x) dx = c \mathbb{P}(g(X) \geq c) \\ &\implies \mathbb{P}(g(X) \geq c) \leq \frac{\mathbb{E}(g(X))}{c}.\end{aligned}$$

□

Corollary 4.1.1. (a) Let $g : [0, \infty) \rightarrow \mathbb{R}$ be a non-negative and strictly increasing function such that $\mathbb{E}(g(X))$ is finite. Then, for any $c > 0$ such that $g(c) > 0$,

$$\mathbb{P}(g(X) \geq c) \leq \frac{\mathbb{E}(g(X))}{c}.$$

(b) Let $r > 0$ and $t > 0$. Then

$$\mathbb{P}(|X| \geq t) \leq \frac{\mathbb{E}(|X|^r)}{t^r}, \quad (\text{Markov's Inequality})$$

provided $\mathbb{E}(|X|^r) < \infty$. In particular

$$\mathbb{P}(|X| \geq t) \leq \frac{\mathbb{E}(|X|)}{t},$$

provided $\mathbb{E}(|X|) < \infty$.

Proof. (a)

$$\begin{aligned}\mathbb{P}(|X| \geq c) &= \mathbb{P}(g(X) \geq g(c)) \quad (\text{since } g \text{ is strictly increasing}) \\ &\leq \frac{\mathbb{E}(g(X))}{g(c)} \quad (\text{by Theorem 4.1}).\end{aligned}$$

(b) Let $g : [0, \infty) \rightarrow \mathbb{R}$ be given by

$$g(x) = x^r, \quad x \geq 0, \quad r > 0.$$

Then g is strictly increasing on $[0, \infty)$ and is non-negative. Using (a) we get

$$\mathbb{P}(|X| \geq t) \leq \frac{\mathbb{E}(g(X))}{g(t)} = \frac{\mathbb{E}(|X|^r)}{t^r}.$$

□

Theorem 4.2 (Chebyshev inequality). *Let X be a RV with finite variance σ^2 and $\mathbb{E}(X) = \mu$. Then, for any $\epsilon > 0$,*

$$\mathbb{P}(|X - \mu| \geq \epsilon\sigma) \leq \frac{1}{\epsilon^2}.$$

Proof. Using the above Corollary

$$\mathbb{P}(|X - \mu| \geq \epsilon\sigma) \leq \frac{\mathbb{E}(|X - \mu|^2)}{\epsilon^2\sigma^2} = \frac{\mathbb{E}((X - \mu)^2)}{\epsilon^2\sigma^2} = \frac{1}{\epsilon^2}.$$

□

Example 4.1. *Let X be a RV with PMF*

$$f(x) = \begin{cases} \frac{1}{8}, & \text{if } x = -1, 1 \\ \frac{3}{4}, & \text{if } x = 0 \\ 0, & \text{Otherwise} \end{cases}$$

Then

$$\mathbb{E}(X^2) = \frac{1}{4}, \quad \text{and} \quad \mathbb{P}(|X| \geq 1) = \frac{1}{4}.$$

Using the Markov inequality

$$\mathbb{P}(|X| \geq 1) \leq \mathbb{E}(X^2) = \frac{1}{4}.$$

Example 4.2. *Let X be a RV with PDF*

$$f(x) = \begin{cases} \frac{1}{2\sqrt{3}}, & \text{if } -\sqrt{3} < x < \sqrt{3} \\ 0, & \text{Otherwise} \end{cases}$$

Then

$$\begin{aligned}\mu &= \mathbb{E}(X) = \int_{-\sqrt{3}}^{\sqrt{3}} x \times \frac{1}{2\sqrt{3}} dx = 0 \\ \sigma^2 &= \mathbb{E}(X^2) = \int_{-\sqrt{3}}^{\sqrt{3}} x^2 \times \frac{1}{2\sqrt{3}} dx = 1\end{aligned}$$

and

$$\mathbb{P}(|X| \geq \frac{3}{2}) = 1 - \int_{-3/2}^{3/2} \frac{1}{2\sqrt{3}} dx = 1 - \frac{\sqrt{3}}{2} = 0.134.$$

Using the Markov inequality

$$\mathbb{P}(|X| \geq \frac{3}{2}) \leq \frac{4}{9}\mathbb{E}(X^2) = \frac{4}{9} = 0.444 \dots$$

Definition 4.1. Let $-\infty \leq a < b \leq \infty$. A function $\Psi : (a, b) \rightarrow \mathbb{R}$ is said to be a convex function if

$$\Psi(\alpha x + (1 - \alpha)y) \leq \alpha\Psi(x) + (1 - \alpha)\Psi(y)$$

for all $x, y \in (a, b)$ and for all $\alpha \in [0, 1]$. The function $\Psi(\cdot)$ is said to be strictly convex if the above inequality is strict.

We state the following theorem without providing its proof.

Theorem 4.3. (a) Let $\Psi : (a, b) \rightarrow \mathbb{R}$ be a convex function. Then Ψ is continuous on (a, b) and is almost everywhere differentiable (that is if D is the set of points where Ψ is not differentiable then D does not contain any interval).

(b) Let $\Psi : (a, b) \rightarrow \mathbb{R}$ be a differentiable function. then Ψ is convex (strictly convex) on (a, b) iff Ψ' is non-decreasing (strictly increasing) on (a, b) .

(c) Let $\Psi : (a, b) \rightarrow \mathbb{R}$ be a twice differentiable function. Then Ψ is convex (strictly convex) on (a, b) iff $\Psi''(x) \geq (>)0, \forall x \in (a, b)$.

Theorem 4.4 (Jensens's Inequality). Let $\Psi : (a, b) \rightarrow \mathbb{R}$ be a convex function and let X be a RV with DF F having support $S \subseteq (a, b)$. Then

$$\mathbb{E}(\Psi(X)) \geq \Psi(\mathbb{E}(X)),$$

provided the expectation exists.

Proof. We provide the prove for special case when Ψ is twice differentiable on (a, b) so that $\Psi''(x) \geq 0, \forall x \in (a, b)$. The Taylor series expansion of $\Psi(x)$ about $\mu = \mathbb{E}(X)$ of order 2:

$$\Psi(x) = \Psi(\mu) + (x - \mu)\Psi'(\mu) + \frac{(x - \mu)^2}{2!}\Psi''(\zeta), \forall x \in (a, b),$$

for some ζ between μ and x . Clearly

$$\begin{aligned} \Psi(x) &\geq \Psi(\mu) + (x - \mu)\Psi'(\mu), \forall x \in (a, b) \\ \implies \mathbb{E}(\Psi(x)) &\geq \mathbb{E}(\Psi(\mu) + (x - \mu)\Psi'(\mu)) \\ &= \Psi(\mu) = \Psi(\mathbb{E}(X)). \end{aligned}$$

□

Example 4.3. (a) For any RV X

$$\mathbb{E}(X^2) \geq (\mathbb{E}(X))^2$$

since $\Psi(x) = x^2, x \in \mathbb{R}$ is a convex function.

(b) For any RV X with $\mathbb{P}(X > 0) = 1$

$$\mathbb{E}(\ln(X)) \geq \ln(\mathbb{E}(X))$$

since $\Psi(x) = -\ln(x),$ is a convex function on $(0, \infty)$.

(c) For any RV X

$$\mathbb{E}(e^X) \geq e^{\mathbb{E}(X)}$$

since $\Psi(x) = e^x, x \in \mathbb{R}$ is a convex function.

(d) For any RV X with $\mathbb{P}(X > 0) = 1$

$$\mathbb{E}(X)\mathbb{E}\left(\frac{1}{X}\right) \geq 1$$

since $\Psi(x) = \frac{1}{x}, x > 0$ is a convex function.

Example 4.4. Let $a_1, \dots, a_n, w_1, \dots, w_n$ be positive constants such that $\sum_{i=1}^n w_i = 1$. prove that the AM, GM, HM inequality

$$\sum_{i=1}^n a_i w_i \geq \prod_{i=1}^n a_i w_i \geq \frac{1}{\sum_{i=1}^n \frac{w_i}{a_i}}.$$

Solution: Let X be a RV with PMF

$$f(x) \begin{cases} w_i, & \text{if } x = a_i, i = 1, \dots, n \\ 0, & \text{Otherwise} \end{cases}$$

Then $\Psi(x) = \ln(x)$, $x > 0$, is a convex function. Therefore

$$\begin{aligned} \mathbb{E}(\Psi(X)) &\geq \Psi(\mathbb{E}(X)) \\ \implies \mathbb{E}(-\ln(x)) &\geq -\ln(\mathbb{E}(x)) \\ \implies -\sum_{i=1}^n (\ln a_i) w_i &\geq -\ln\left(\sum_{i=1}^n a_i w_i\right) \\ \implies \ln\left(\sum_{i=1}^n a_i w_i\right) &\geq \ln\left(\prod_{i=1}^n a_i^{w_i}\right) \\ \implies \sum_{i=1}^n a_i w_i &\geq \prod_{i=1}^n a_i^{w_i}. \end{aligned}$$

Replacing a_i 's by $\frac{1}{a_i}$'s we get

$$\prod_{i=1}^n a_i^{w_i} \geq \sum_{i=1}^n \frac{w_i}{a_i}.$$

Therefore

$$\sum_{i=1}^n a_i w_i \geq \prod_{i=1}^n a_i w_i \geq \frac{1}{\sum_{i=1}^n \frac{w_i}{a_i}}.$$